

ELEC 3200 Final Exam Review

Review Session

Final exam: 2026-05-20 · 12:30–15:30

Today's Focus

- Cumulative review, but **focused on later-semester topics**
- Earlier-semester material (modeling, block diagrams, Routh stability criterion, pole placement) — review midterm slides

Today's main topics:

- System type & steady-state error
- Integral control & tradeoffs
- Second-order response & dominant poles
- Root locus + Routh

You Should Already Be Able To...

- Model electrical systems: KCL/KVL, resistor/inductor/capacitor laws
- Model rotational mechanics: torque balance, damping, gravity torque, moment of inertia
- Recognize electromechanical coupling: motor torque, back-EMF
- Build a state-space model: $\dot{x} = Ax + Bu$, $y = Cx + Du$
- Convert to transfer function: $G(s) = C(sI - A)^{-1}B + D$
- Simplify a block diagram: $T(s) = \frac{G}{1+GH}$
- Use the Routh table: first-column sign test $\rightarrow$ parameter range
- Pole placement: solve $a(s)p(s) + b(s)q(s) = c(s)$ for desired characteristic polynomial

If any of these feel weak — review the midterm slides.

Pole Placement — Counting Equations

Use the controller parameterization stated in the problem. For the first-order form used in the course:

$$C(s) = \frac{q_0 s + q_1}{p_0 s + p_1}$$

with plant $P(s) = b(s)/a(s)$ of degree 2, the closed-loop characteristic polynomial

$$a(s)p(s) + b(s)q(s) = c(s)$$

has degree 3. Match coefficients:

- Treat the displayed coefficients (p_0, p_1, q_0, q_1) as unknowns
- 4 coefficients of $c(s) \rightarrow$ normalize $p_0 = 1 \rightarrow$ **3 free unknowns, 3 equations** ✓

Rule of thumb: for a degree- n plant and a degree- m controller, the closed-loop char poly has degree $n + m$. Follow the given controller form; do not invent or delete coefficients.

System Type & Steady-State Error

- **Definition:** System type = number of poles of $L(s)$ at the origin.
- **Setup:** Unity feedback, $L(s) = C(s)P(s)$.

$$E(s) = \frac{R(s)}{1+L(s)}, e_{\infty} = \lim_{s \rightarrow 0} s E(s)$$

Input	Type 0	Type 1	Type 2
Step	$\frac{1}{1 + K_p}$	0	0
Ramp	∞	$\frac{1}{K_v}$	0
Parabola	∞	∞	$\frac{1}{K_a}$

Stable $\neq$ zero error. Stability is necessary, not sufficient.

Error Constants

$$K_p = \lim_{s \rightarrow 0} L(s) \quad K_v = \lim_{s \rightarrow 0} s L(s) \quad K_a = \lim_{s \rightarrow 0} s^2 L(s)$$

- All limits use $L(s)$ — the **open-loop** loop gain — **not** the closed-loop $T(s)$
- FVT is valid **only when the closed-loop is stable**
- If $sE(s)$ has any RHP pole $\rightarrow$ FVT is invalid

Integral Control: The Tradeoff

$$C(s) = \frac{K_I}{s}$$

- Adds a pole at the origin
- Usually **raises system type by 1** → improves tracking
- In the plants we are considering, it can reduce damping ratio ζ

Typical tradeoff: better tracking may come with worse damping. When the resulting second-order response is underdamped, larger K_I usually means larger overshoot. Always compute the closed-loop poles to check.

Example — Proportional vs Integral

Plant: $P(s) = \frac{4}{s+3}$, unity feedback.

Controller A: $C(s) = \frac{3}{s}$

Find:

1. System type 0
2. Steady-state error to a step 0
3. Steady-state error to a ramp

$$\lim_{s \rightarrow 0} \frac{sR(s)}{1+CP}$$



Board work

$$\begin{aligned} \lim_{s \rightarrow 0} \frac{1}{1 + \frac{12}{s(s+3)}} \cdot \frac{1}{s} \\ = \lim_{s \rightarrow 0} \frac{(s+3)}{s(s+3)+12} = \frac{1}{4} \end{aligned}$$

Common Mistakes — Steady-State Error

-  Computing K_p, K_v on the **closed-loop** TF instead of $L(s)$
-  Confusing $\frac{1}{1+L}$ with $E(s)$ — must multiply by $R(s)$ first
-  Assuming “stable $\Rightarrow$ zero error” — false for Type 0 + step
-  Applying FVT when $sE(s)$ has RHP poles — **invalid**

Second-Order — Standard Form

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

From a denominator $s^2 + as + b$:

$$\omega_n = \sqrt{b}, \quad \zeta = \frac{a}{2\sqrt{b}}$$

Don't confuse ω_n^2 with ω_n — off by a square root!

Second-Order — Performance Specs

Spec	Formula
Damped frequency	$\omega_d = \omega_n \sqrt{1 - \zeta^2}$
Percent overshoot	$\%OS = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%$
Peak time	$t_p = \pi/\omega_d$
Settling time (n%)	$e^{-\sigma t_s} \leq n/100 \quad t_s \approx -\ln(\frac{n}{100})/\sigma$
Rise time	$t_r \approx 1.8/\omega_n$

$$M_p = \exp\left(-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}\right)$$

Dominant Poles

A high-order system can be approximated by a **second-order** system using only its dominant poles.

Rule of thumb:

$$\frac{D}{d} \geq 5 \sim 10 \quad \Rightarrow \quad \text{approximation valid}$$

where:

- d = distance of dominant pair from imaginary axis
- D = distance of next-closest pole

When a question says “based on the dominant poles” → use this approximation, but always check the ratio first.

Example — Dominant-Pole Approximation

Closed-loop poles: $\{-8, -1 \pm j\}$

Find ω_n, ζ using the dominant-pole approximation.

Steps:

1. Identify the dominant pair; check the ratio
2. Build $(s - p_1)(s - p_2)$ and expand
3. Match standard form $\rightarrow$ read off ω_n, ζ



Board work

Common Mistakes — Second-Order

-  Using **open-loop** denominator instead of **closed-loop** for ω_n, ζ
-  ω_n^2 vs ω_n — square-root error
-  Applying 2nd-order formulas to high-order systems without checking the dominance ratio
-  Using a settling-time constant without checking the requested percentage band

Root Locus — Sketching Checklist

Rule A: number of branches = number of open-loop poles

Rule B: branches start at open-loop poles $p_1, \dots, p_n$

Rule C: branches end at open-loop zeros $z_1, \dots, z_m$ or ∞

Rule D: real-axis locus: # real poles + zeros to the right is **odd**

Rule E: asymptotes:

– angles: $\theta_q = \frac{(2q+1) 180^\circ}{n-m}$

Rule F: $j\omega$ -axis crossings $\rightarrow$ **use the Routh table**

Trick — Get into $1 + KG(s) = 0$ Form

If your characteristic equation has the form

$$a(s) + K \cdot b(s) = 0$$

just divide through:

$$1 + K \cdot \frac{b(s)}{a(s)} = 0 \quad \Rightarrow \quad G(s) = \frac{b(s)}{a(s)}$$

Mini exercise: $s^2 + 6s + K = 0$

$$\rightarrow a(s) = s^2 + 6s, \quad b(s) = 1 \quad \Rightarrow \quad G(s) = \frac{1}{s(s+6)}$$

Optional Example — Routh K-Range

Loop transfer function:

$$L(s) = \frac{1}{s(s+2)(s+4)}, \quad \text{unity feedback}$$

Find the range of K for closed-loop stability.

Plan: 1. Form the closed-loop characteristic equation 2. Build the Routh table 3. Apply first-column-positive conditions $\rightarrow$ range of K



Board work if time · Sketching gives the shape; Routh gives the threshold.

Common Mistakes — Root Locus

-  Plotting from **closed-loop** poles instead of **open-loop**
-  Forgetting **repeated** poles (e.g., double integrator → two branches at origin)
-  Clearly mark poles(X) and zeros(O)
-  Reporting locus shape but not giving the **gain range** when asked

Quick Tips for the Exam

- **Read the criterion** — Use the formula the question gives.
- **Use the formula stated** — e.g. $t_r \approx 1.8/\omega_n$ if the question states it
- **Decimal places** — keep exactly what the question asks for
- **Pole placement** — count equations: $\deg(\text{plant}) + \deg(\text{controller}) = \deg(\text{char poly})$
- **Open vs closed loop** — error constants from $L(s)$; stability/transient from closed-loop char eq
- **Linearization** — drop constants at equilibrium, $\sin\theta \approx \theta$
- **Stuck on a block diagram?** Label every signal, write equations
- **FVT** valid only when the closed-loop is stable

Before the Exam, Make Sure You Can...

- ✓ Write $E(s) = \frac{R(s)}{1+L(s)}$ and apply FVT
- ✓ Read system type from $L(s)$
- ✓ Convert a 2nd-order denominator to ω_n, ζ
- ✓ Apply dominant-pole approximation (and verify the ratio)
- ✓ Sketch root locus and use Routh for the stable K range